NHDF EDGE - MEASURED PRACTICAL INTERPRETATION

What 102.37 tokens/s means in practice

Qwen3-30B-A3B on an RTX 5070 Ti Laptop GPU (12 GB), through the NHDF hybrid substrate

MEASURED BENCHMARK - NOT A GUARANTEED CHAT RATE

STEADY GENERATION MICROBENCHMARK

102.37 **tokens/s**

64-token llama-bench, mean of 3 samples

500 output tokens: about 4.9 s **1,000: about 9.8 s**

HUMAN-SCALE TRANSLATION

~77 words/s

Using the rough English conversion of 0.75 words per token. Token-to-word ratios vary by language and content.

If sustained, output arrives faster than normal reading.

ROUGH INTERACTIVE-USE HEURISTIC

How generation speed usually feels

102.37 measured

2-5	5-10	10-20	20-40	40+
frustrating	patient	comfortable	feels live	faster than reading

END-TO-END WAIT COMPONENTS

- **458.53 tok/s** prompt processing
  A 1,000-token input is about 2.2 s at this microbenchmark rate.
- **15.00-23.96 s** cold model load
  Observed startup range. This is largely avoided while the model remains resident.

Cold response time = load + prompt processing + generation.

THE IMPORTANT LIMIT

102.37 tok/s is a 64-token llama-bench microbenchmark. It is not yet a measured full chat session or a guaranteed rate.

Sampling, UI overhead, prompt length, context occupancy, power limits, and thermals can reduce real speed. The 25.34-28.83 tok/s exact-output CLI observations generated only one token each, so they are not sustained-rate estimates.

WHAT IS ACTUALLY ESTABLISHED

Locally usable now; sustained chat UX is the next measurement.

- Complete 30.532B-parameter MoE model; about 3.3B parameters active per token.
- 49/49 layers offloaded; allocated-8K peak 10,487 MiB with 1,740 MiB headroom.
- 4/4 functional prompts passed with correct, meaningful output - not garbage data.

Next decisive test: warm time-to-first-token plus 256-1,000 generated tokens
at short and genuinely filled long contexts, with peak VRAM and thermals recorded

Evidence: metrics/local/gguf_iq2m_llama_bench.json and packs/.../evidence/functional_gate.json

Runtime: llama.cpp b6014 | Payload codec: GGUF/IQ2_M | Substrate: NHDF v0.3 hybrid